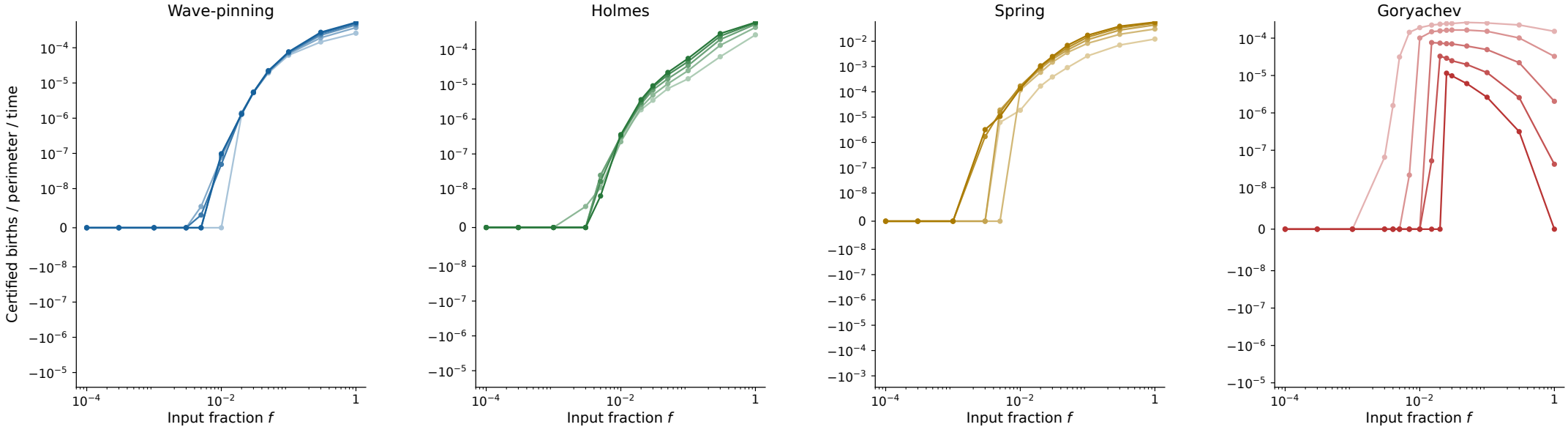
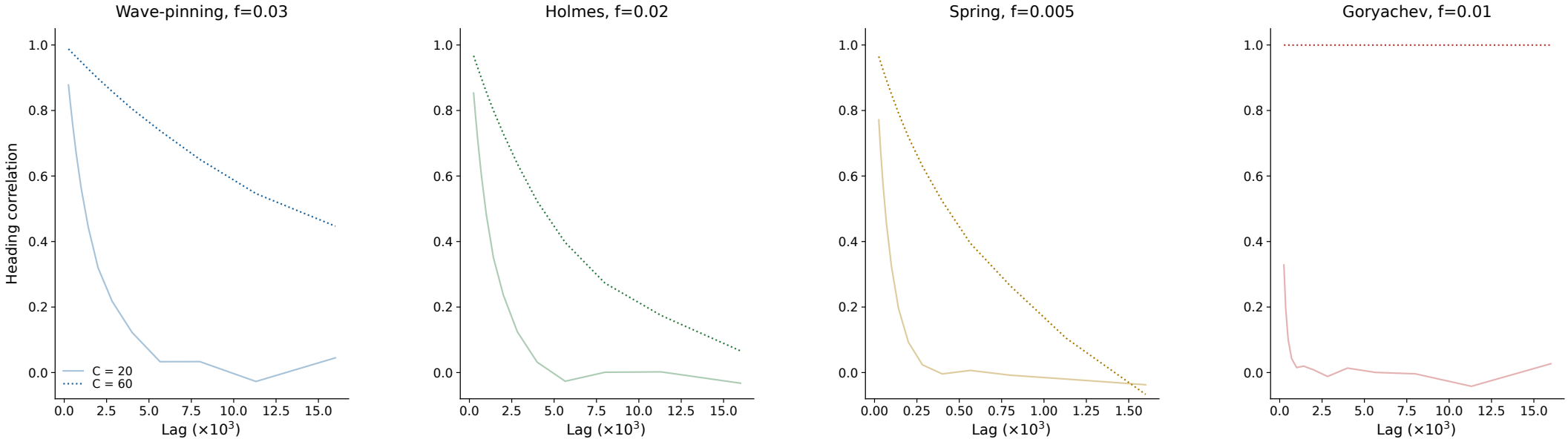
S4.1 Certified persistent-secondary birth rates



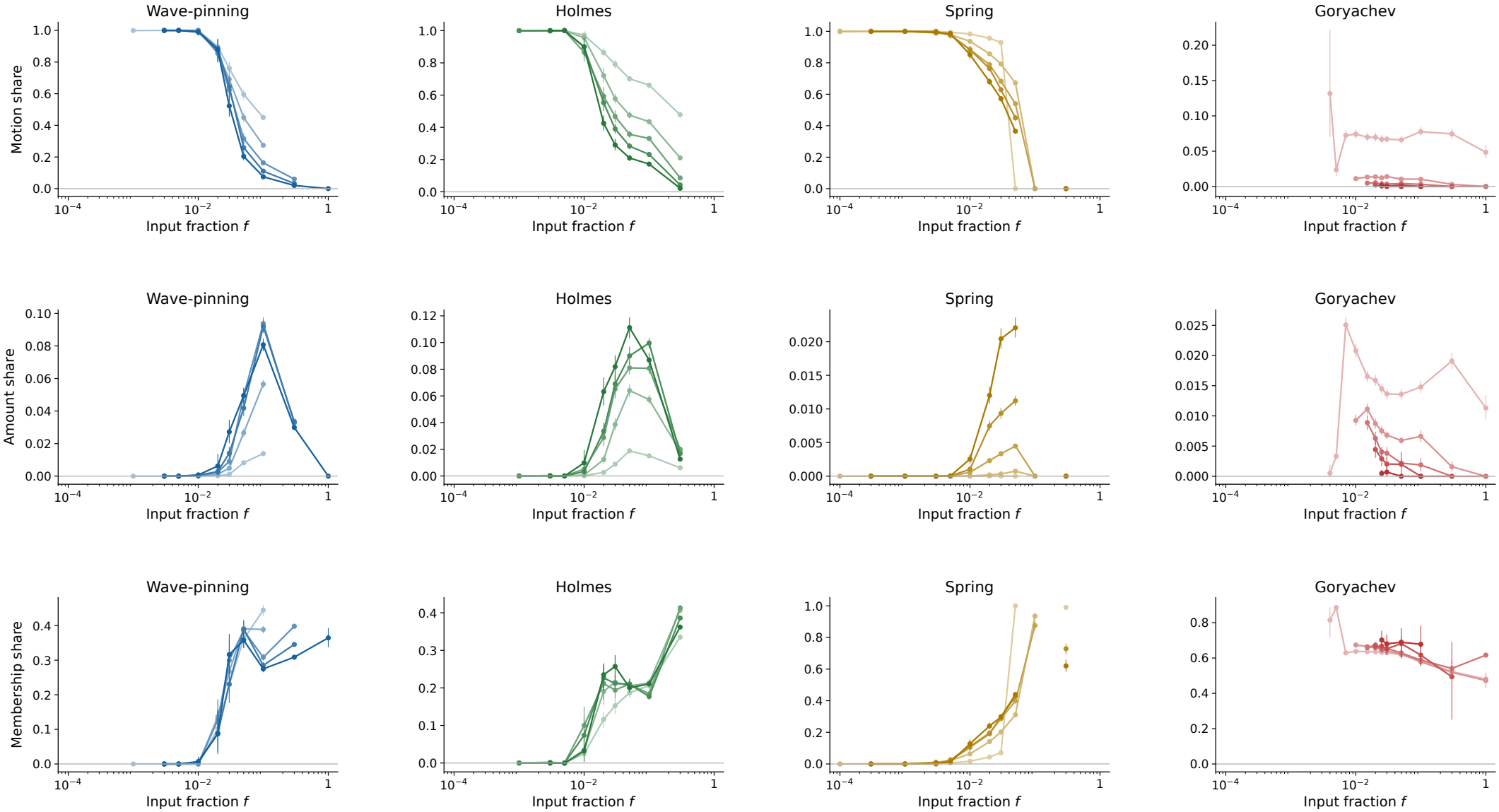
Rates are certified lower bounds per actual perimeter–time. Uncertified onsets can contain split descendants; native local parameters are unchanged.

S4.1b Native heading correlations at the transition input



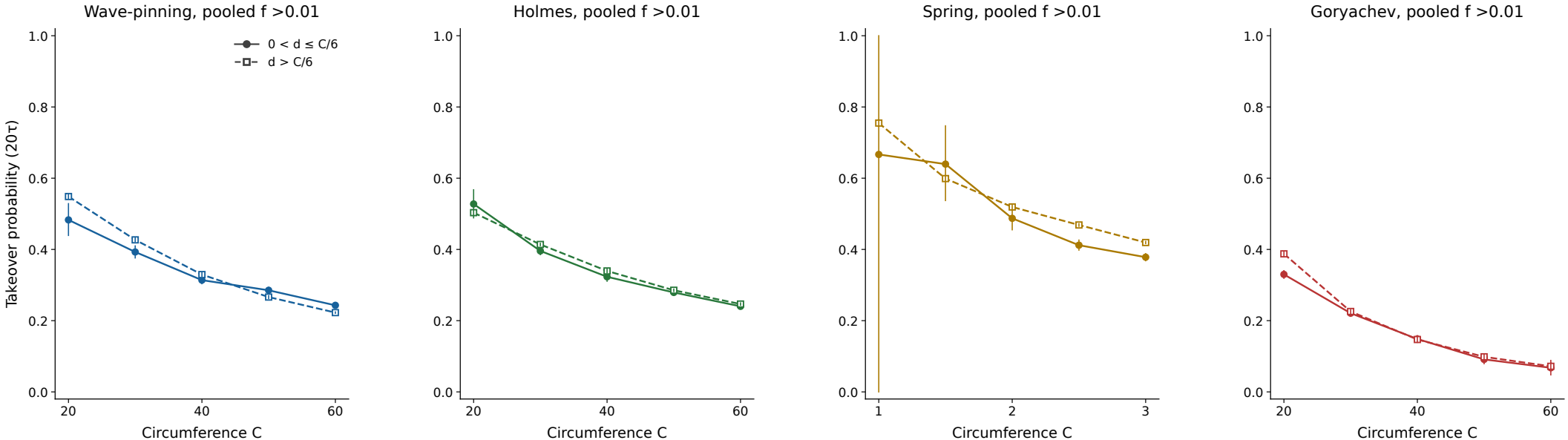
Superseded round 5 model heading-correlation row, at the unchanged selected input. These are the complete independent-history estimates; source CSV condition_lag_metrics.csv.

S4.2 Motion, amount and membership shares



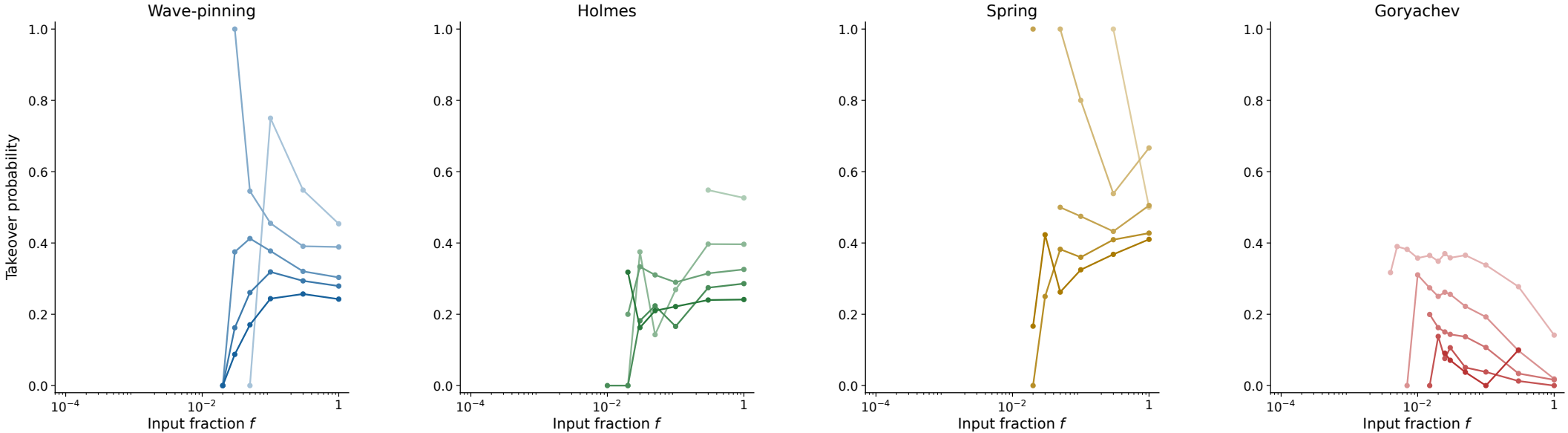
Round5 native-amplitude symmetric Shapley convention. Signed fractions can be negative or exceed one; no change to source profiles, eligibility or events.

S4.3 Pooled finite-horizon takeovers



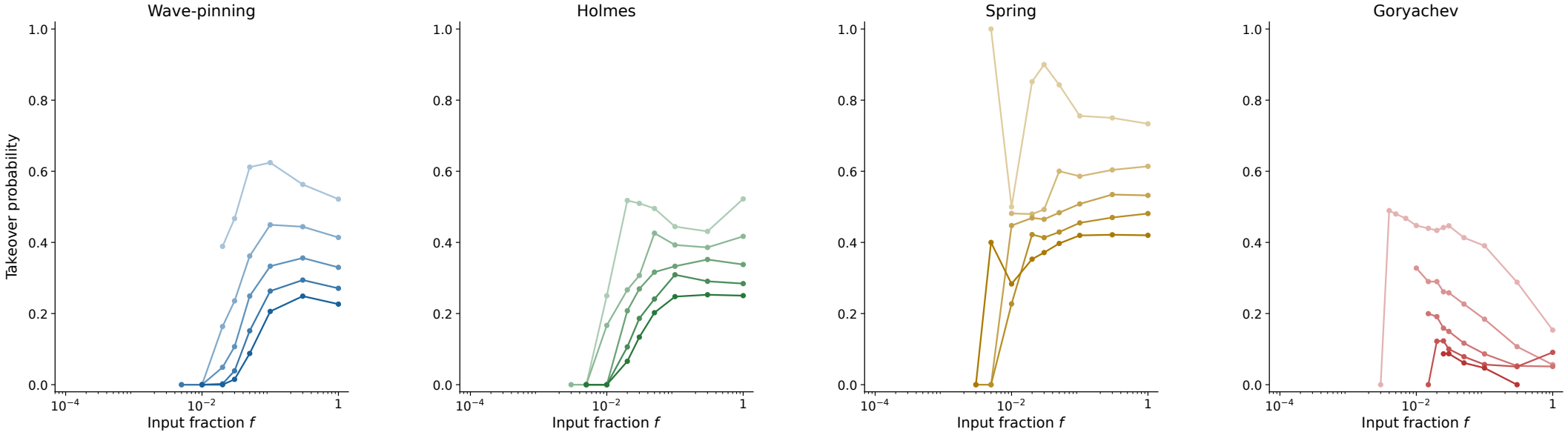
Historical pooled f >0.01 diagnostic, shown only for positive distance. Main I uses a separately selected input per model and >=200 births at every size.

S4.4 Takeover probability at every input: 0 < d ≤ C/6



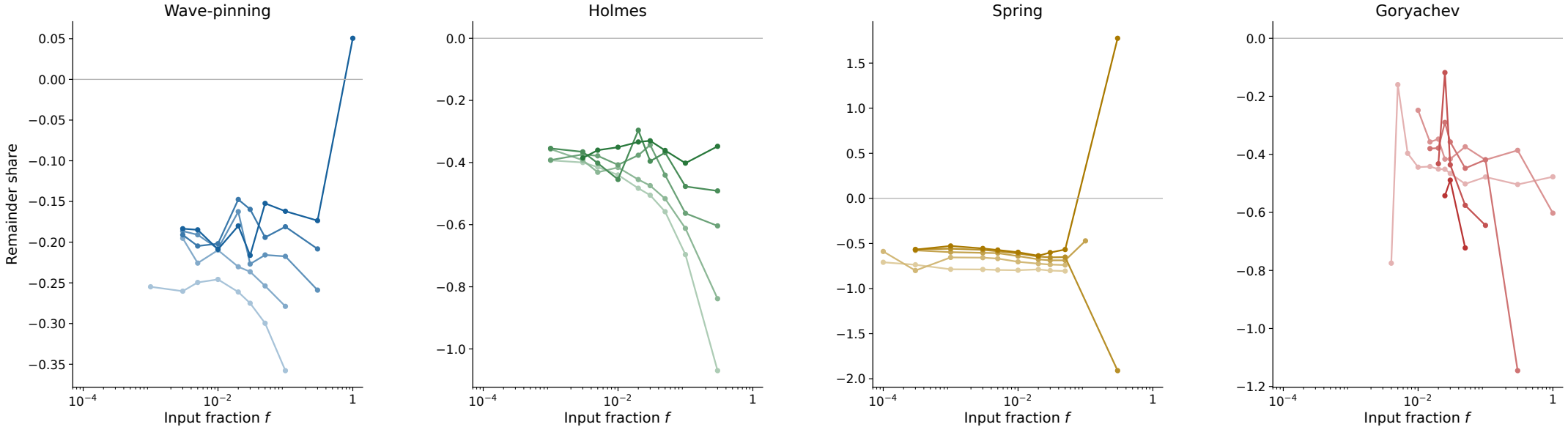
Common follow-up20τ; each curve is one size. Certified no-overlap subset only. Inputs with no certified births remain missing.

S4.4 Takeover probability at every input: d > C/6



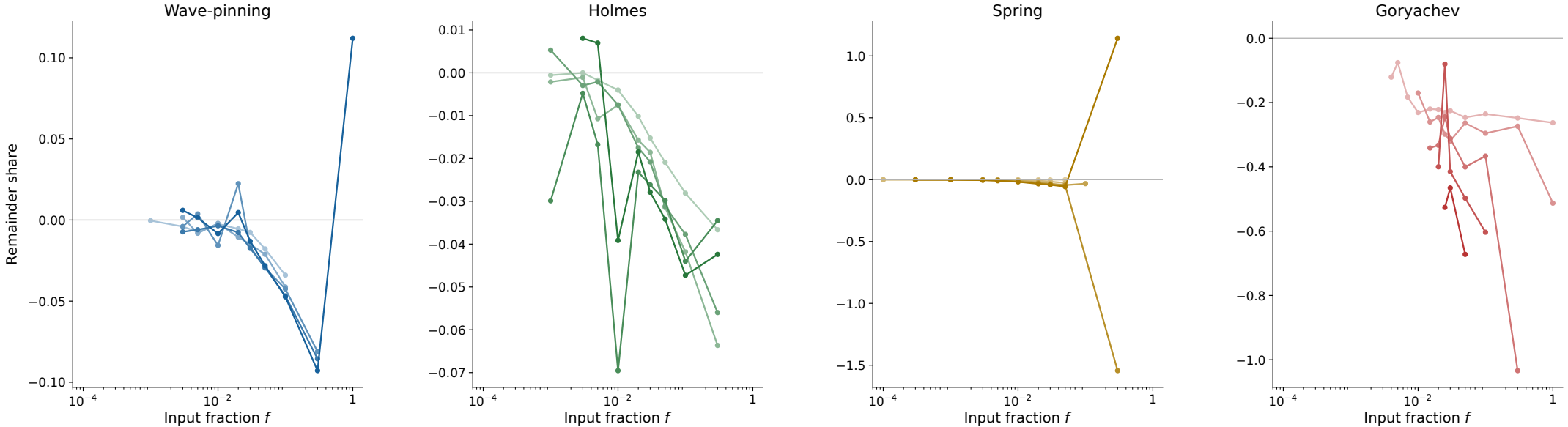
Common follow-up20τ; each curve is one size. Certified no-overlap subset only. Inputs with no certified births remain missing.

S4.5 Own-mask moment check: point matched mass



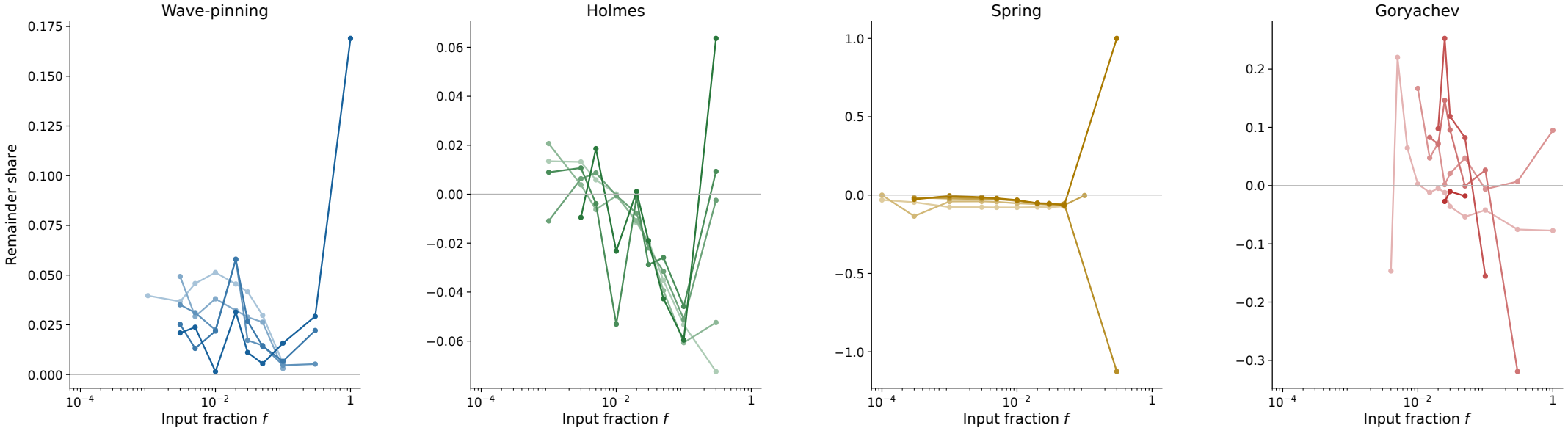
Existing seeds 0/1; saved binned float16 profiles. Native-mass denominator in first 3 variants; last 2 retain round 5 region normalization and native amplitude scaling. Shape enters amount/shape in first 3 and motion/shape in last 2. Algebraic reconstruction is not causal validation.

S4.5 Own-mask moment check: exact full mask



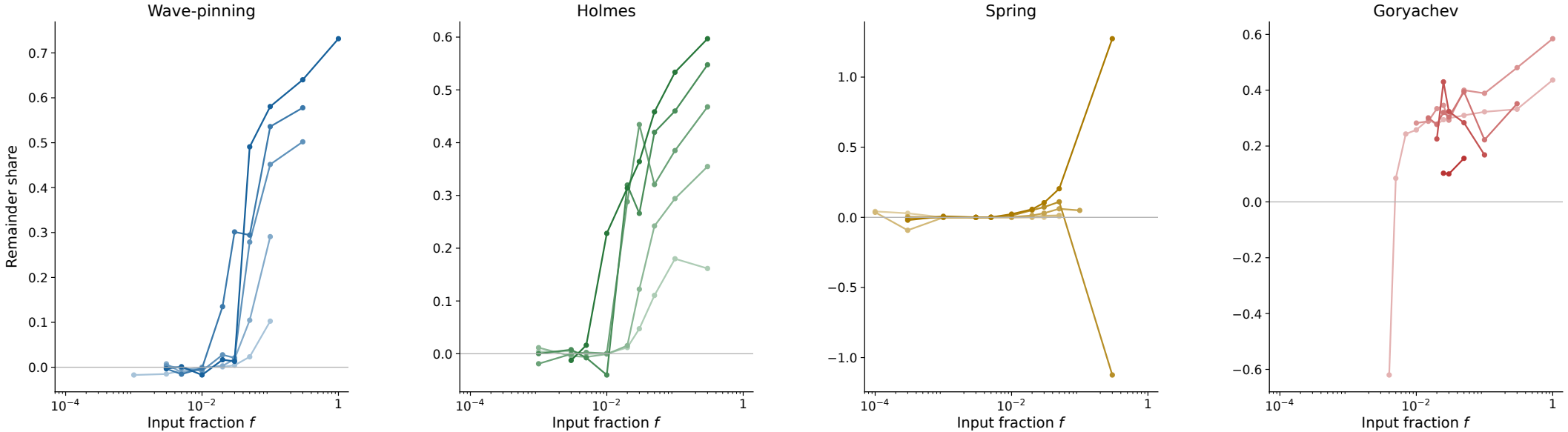
Existing seeds 0/1; saved binned float16 profiles. Native-mass denominator in first 3 variants; last 2 retain round 5 region normalization and native amplitude scaling. Shape enters amount/shape in first 3 and motion/shape in last 2. Algebraic reconstruction is not causal validation.

S4.5 Own-mask moment check: exact contour mask



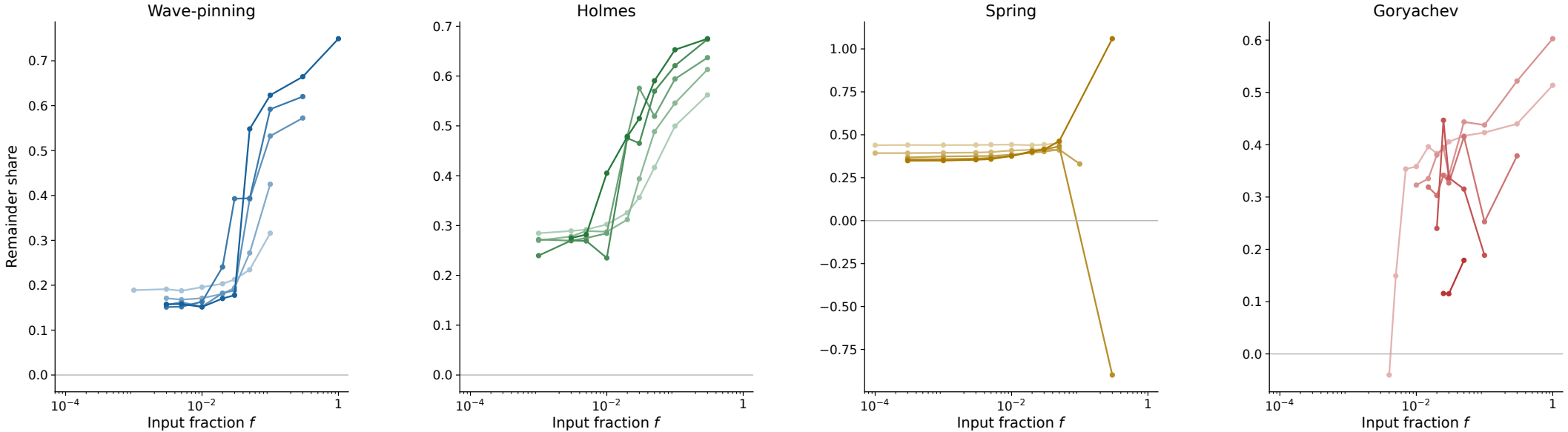
Existing seeds 0/1; saved binned float16 profiles. Native-mass denominator in first 3 variants; last 2 retain round 5 region normalization and native amplitude scaling. Shape enters amount/shape in first 3 and motion/shape in last 2. Algebraic reconstruction is not causal validation.

S4.5 Own-mask moment check: round5 point native amplitude



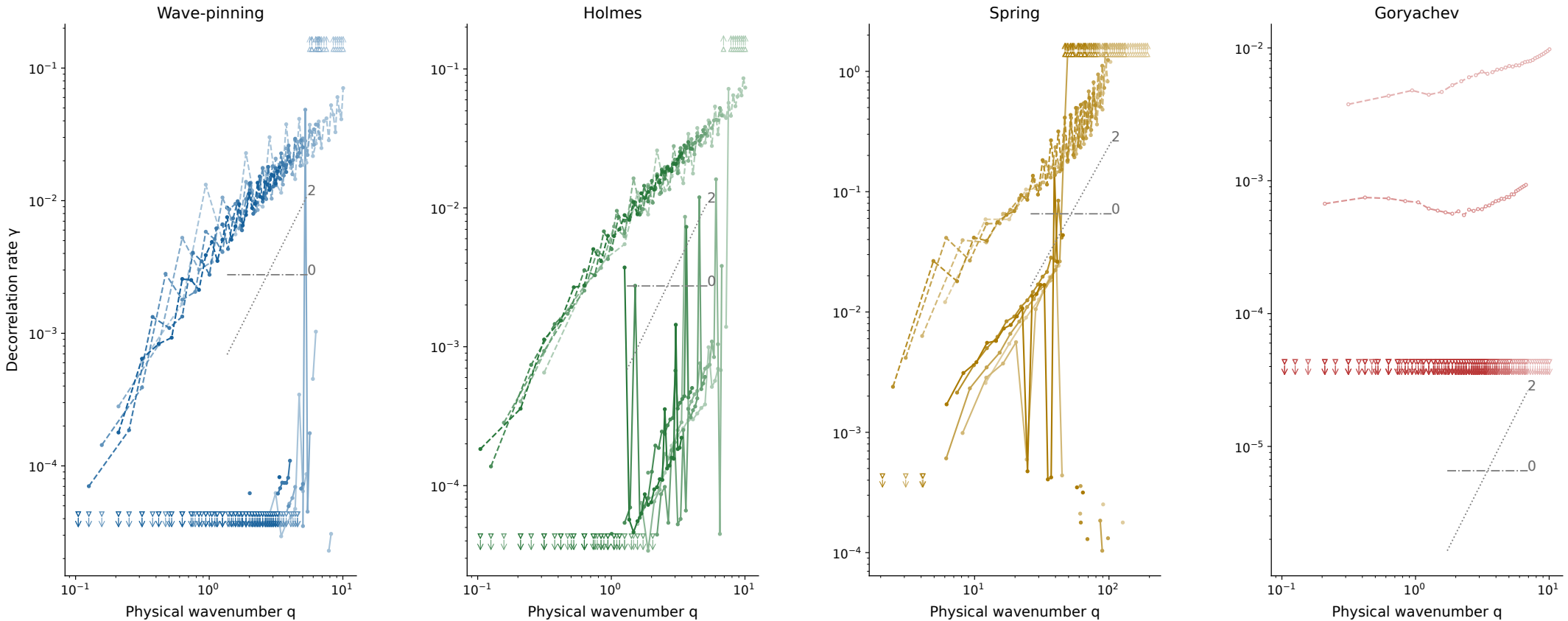
Existing seeds 0/1; saved binned float16 profiles. Native-mass denominator in first 3 variants; last 2 retain round 5 region normalization and native amplitude scaling. Shape enters amount/shape in first 3 and motion/shape in last 2. Algebraic reconstruction is not causal validation.

S4.5 Own-mask moment check: round5 exact native amplitude



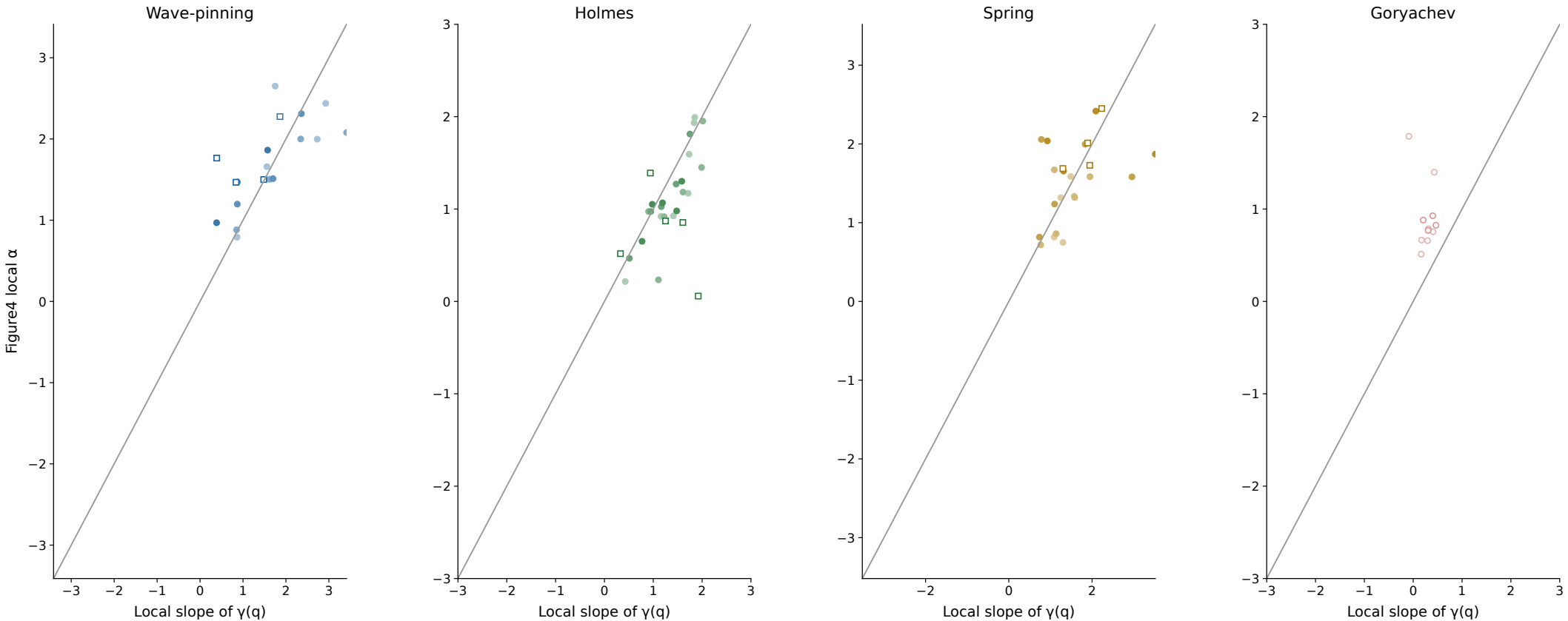
Existing seeds 0/1; saved binned float16 profiles. Native-mass denominator in first 3 variants; last 2 retain round 5 region normalization and native amplitude scaling. Shape enters amount/shape in first 3 and motion/shape in last 2. Algebraic reconstruction is not causal validation.

J Fourier-mode decorrelation versus physical wavenumber



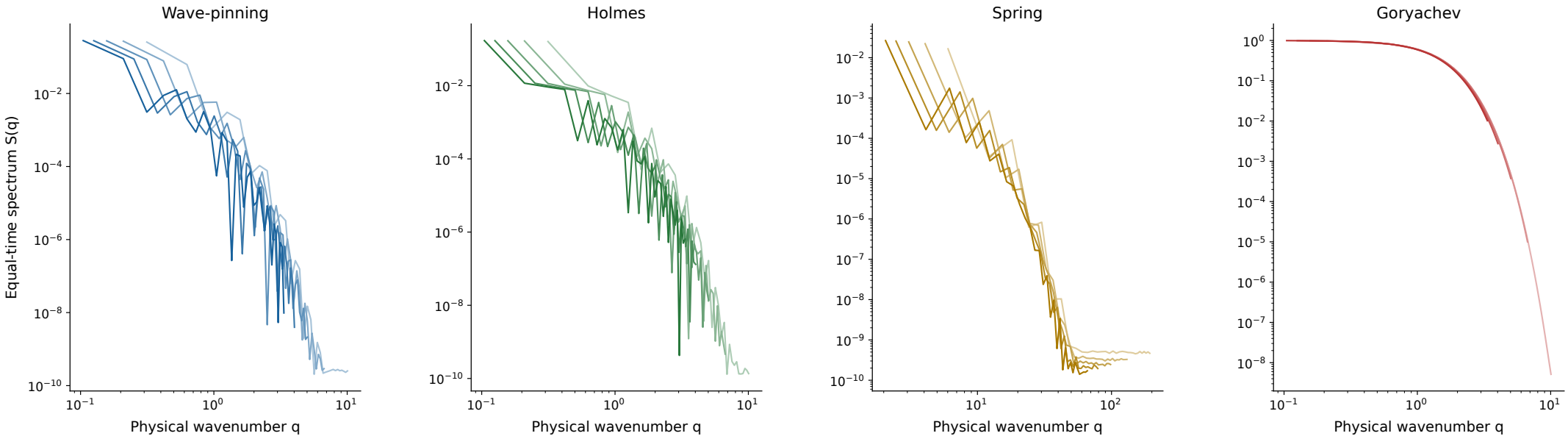
Solid weak f=0.001 (existing exemplars); dashed round 5 transition input. Open triangles with arrows bound rates when half crossing is outside the saved lag range. Targeted records replace strong curves after precision analysis.

K Size exponent versus independently predicted local mode slope



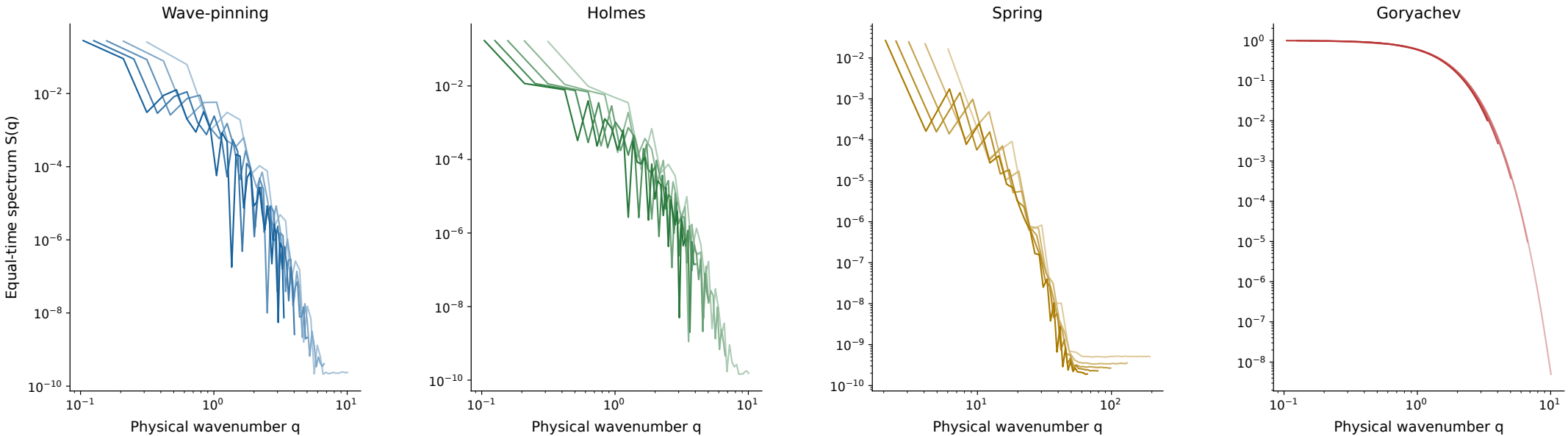
Mode slope uses the nearest 8 mode points from other sizes at the same model and input; the target size is excluded. One-sided predictions are open squares. Forced conditions have open circles. This tests collapse rather than replotting the same size slope.

S4.6 Equal-time spectrum, f = 0.0001



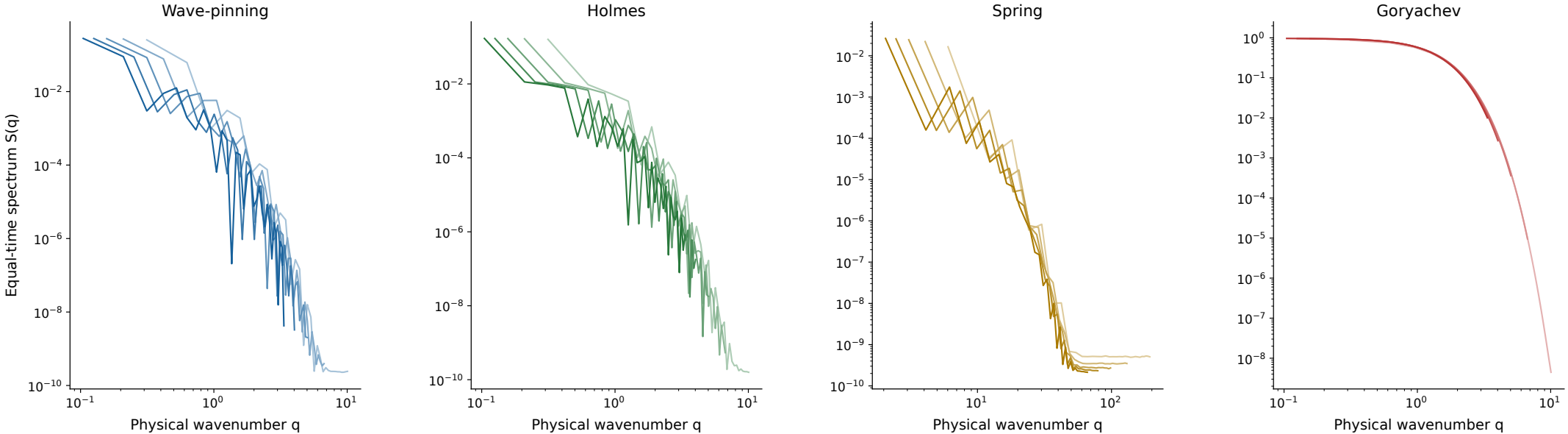
$S(q) = \langle |a_n|^2 \rangle / \langle a_0^2 \rangle$; n=1–32, two existing exemplars/condition. Spring q uses mean C. n* and C/n* are in exemplar_rates.csv.

S4.6 Equal-time spectrum, f = 0.0003



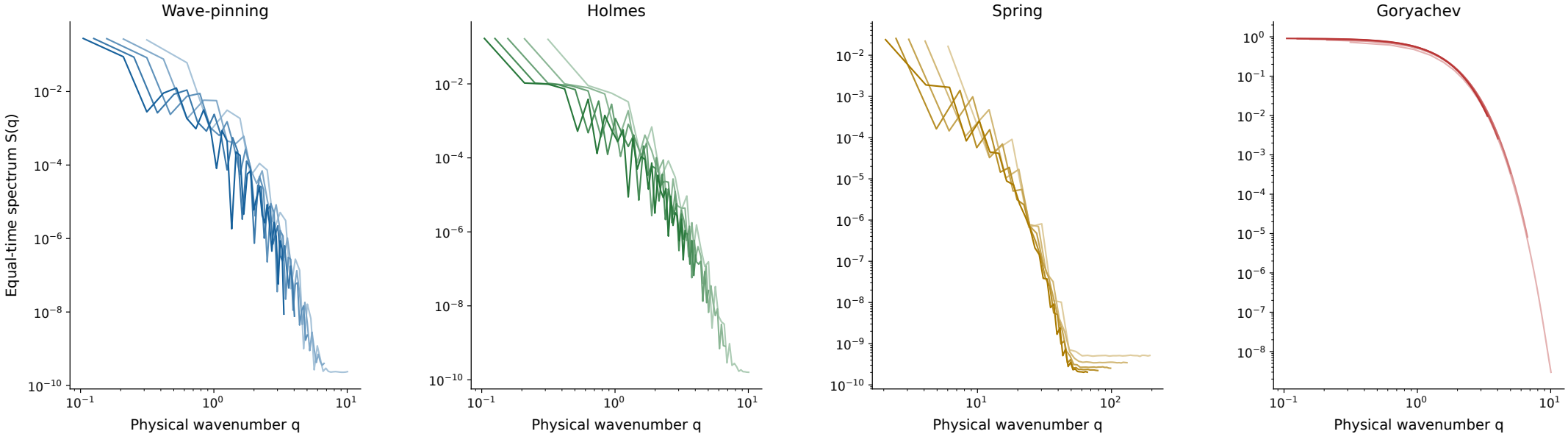
$S(q) = \langle |a_n|^2 \rangle / \langle a_0^2 \rangle$; n=1–32, two existing exemplars/condition. Spring q uses mean C. n* and C/n* are in exemplar_rates.csv.

S4.6 Equal-time spectrum, f = 0.001



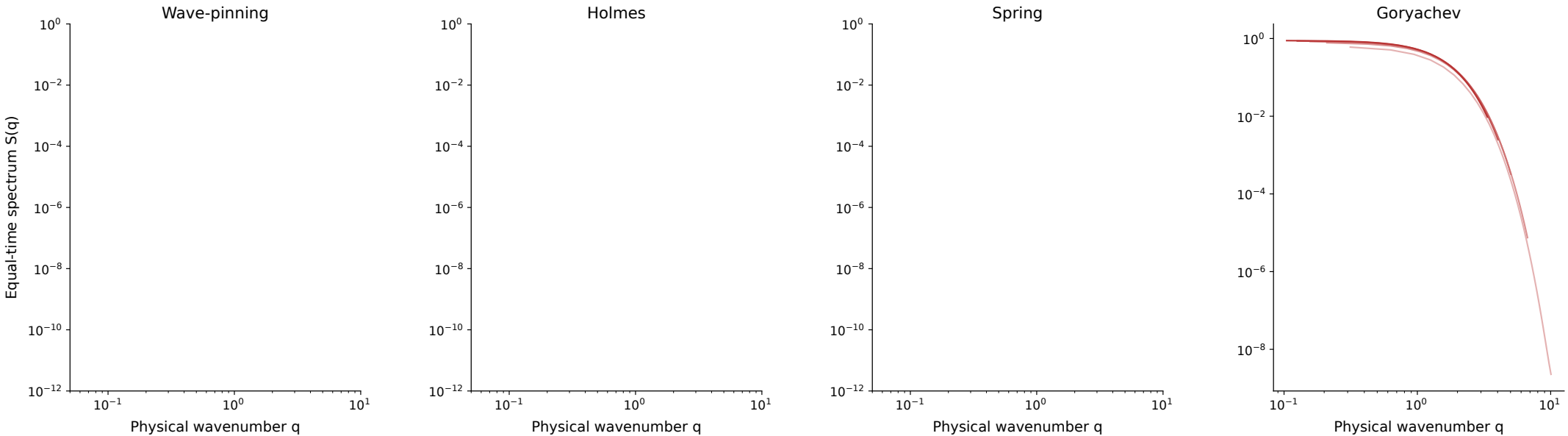
$S(q) = \langle |a_n|^2 \rangle / \langle a_0^2 \rangle$; n=1–32, two existing exemplars/condition. Spring q uses mean C. n* and C/n* are in exemplar_rates.csv.

S4.6 Equal-time spectrum, f = 0.003



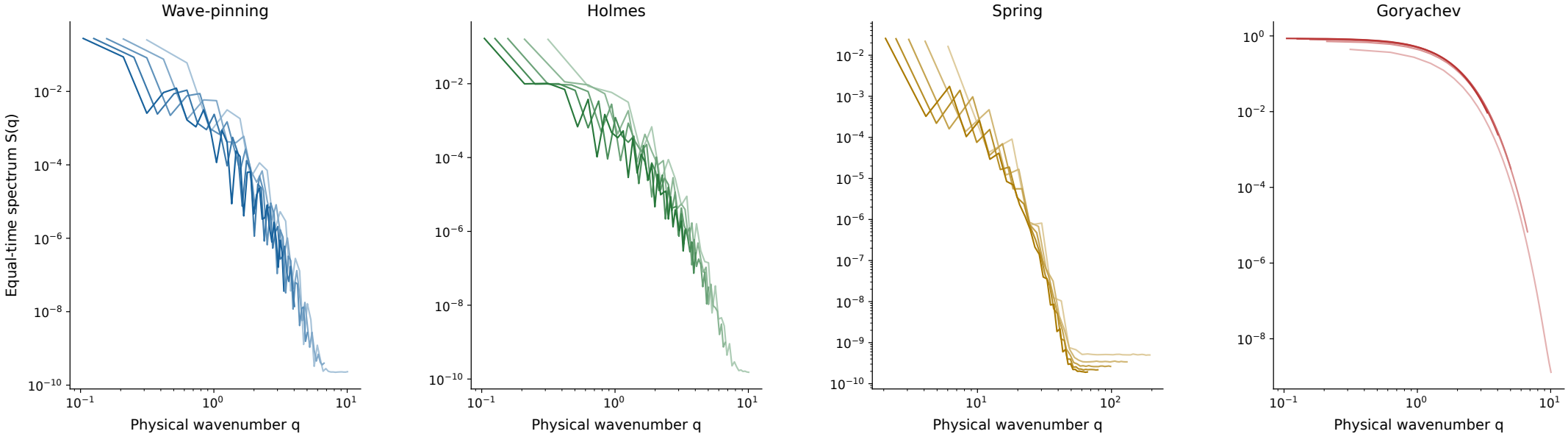
$S(q) = \langle |a_n|^2 \rangle / \langle a_0^2 \rangle$; n=1–32, two existing exemplars/condition. Spring q uses mean C. n* and C/n* are in exemplar_rates.csv.

S4.6 Equal-time spectrum, f = 0.004



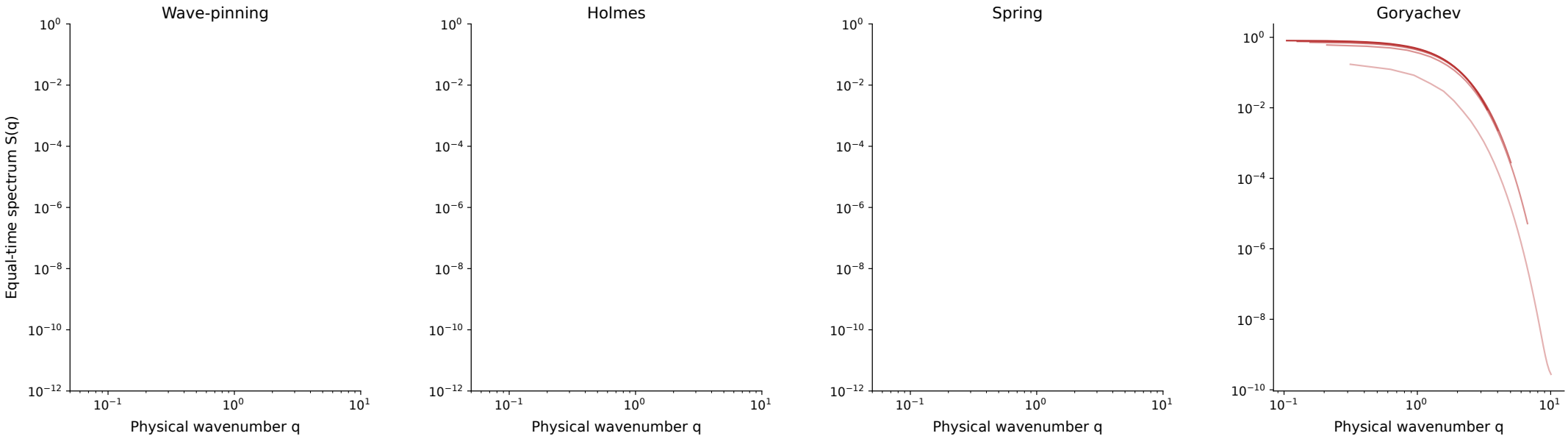
$S(q) = \langle |a_n|^2 \rangle / \langle a_0^2 \rangle$; n=1–32, two existing exemplars/condition. Spring q uses mean C. n* and C/n* are in exemplar_rates.csv.

S4.6 Equal-time spectrum, f = 0.005



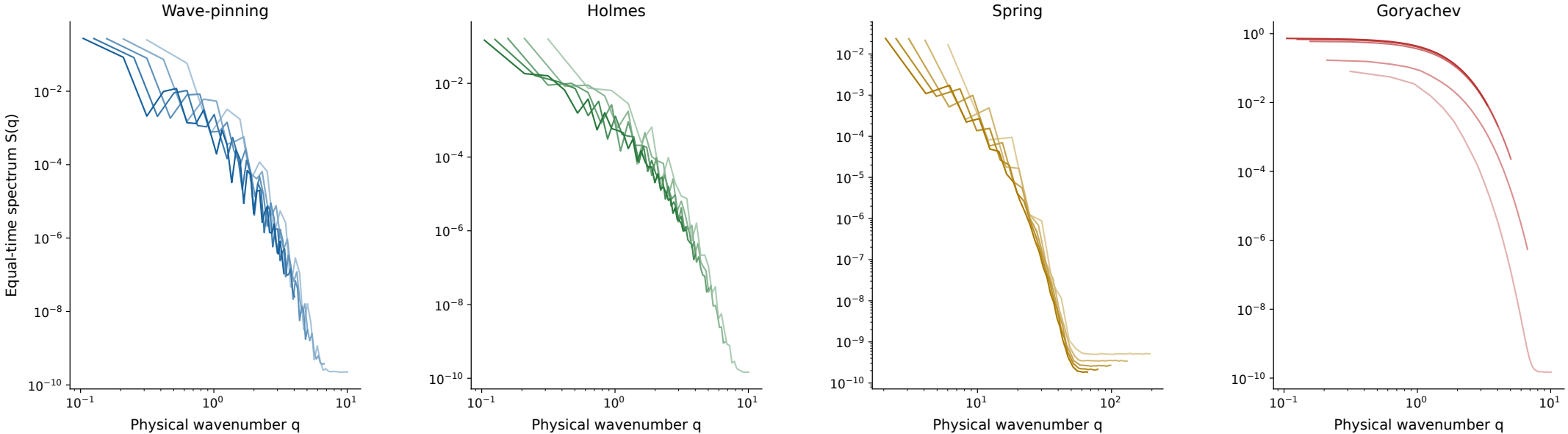
$S(q) = \langle |a_n|^2 \rangle / \langle a_0^2 \rangle$; n=1–32, two existing exemplars/condition. Spring q uses mean C. n* and C/n* are in exemplar_rates.csv.

S4.6 Equal-time spectrum, f = 0.007



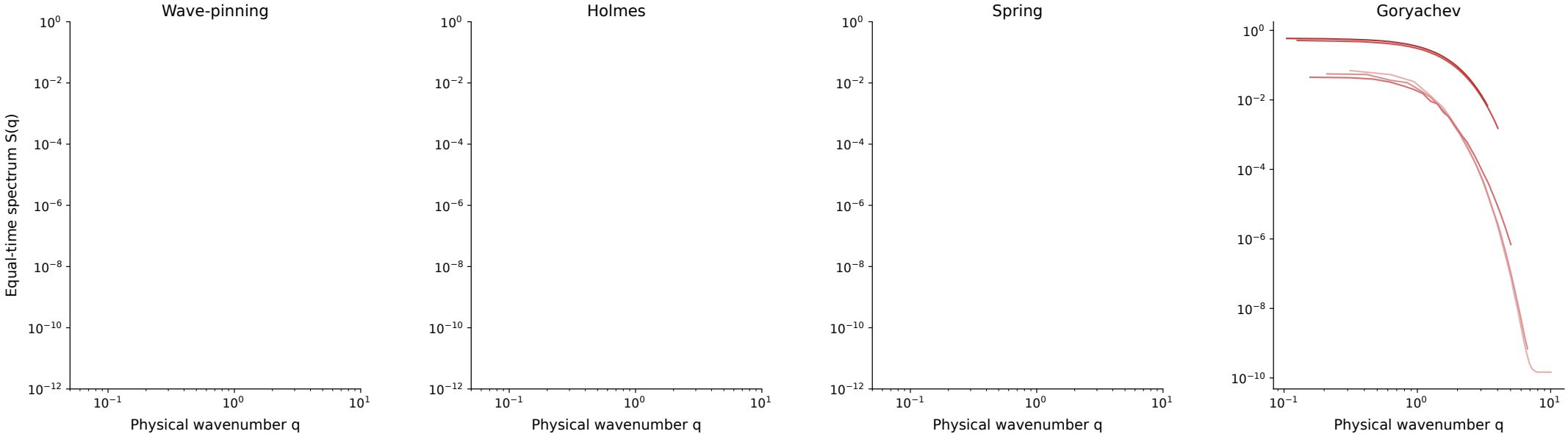
$S(q) = \langle |a_n|^2 \rangle / \langle a_0^2 \rangle$; n=1–32, two existing exemplars/condition. Spring q uses mean C. n* and C/n* are in exemplar_rates.csv.

S4.6 Equal-time spectrum, f = 0.01



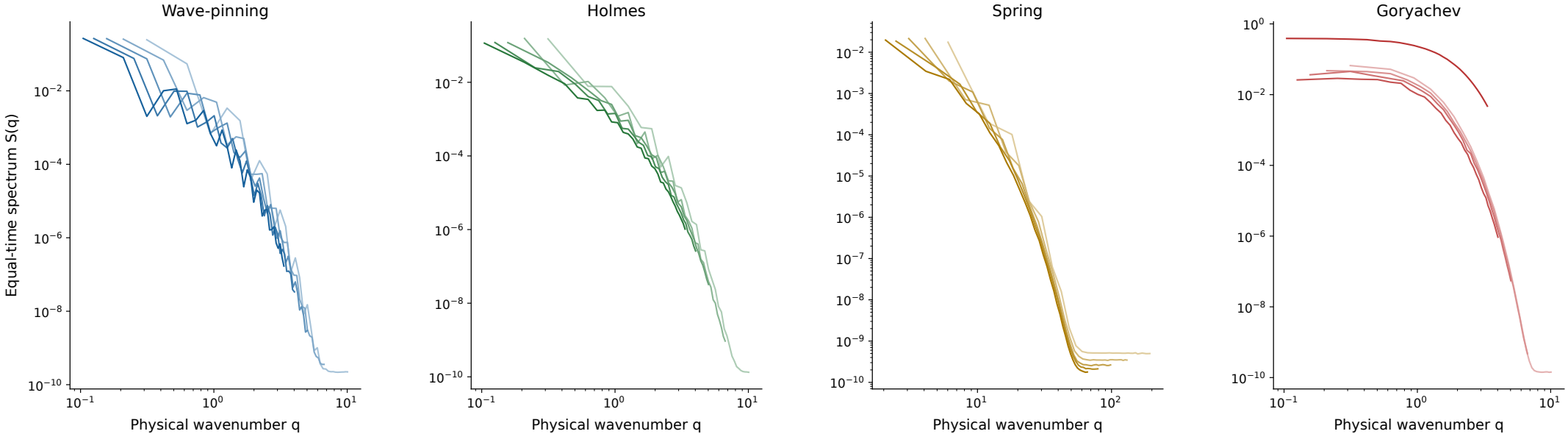
$S(q) = \langle |a_n|^2 \rangle / \langle a_0^2 \rangle$; n=1–32, two existing exemplars/condition. Spring q uses mean C. n* and C/n* are in exemplar_rates.csv.

S4.6 Equal-time spectrum, f = 0.015



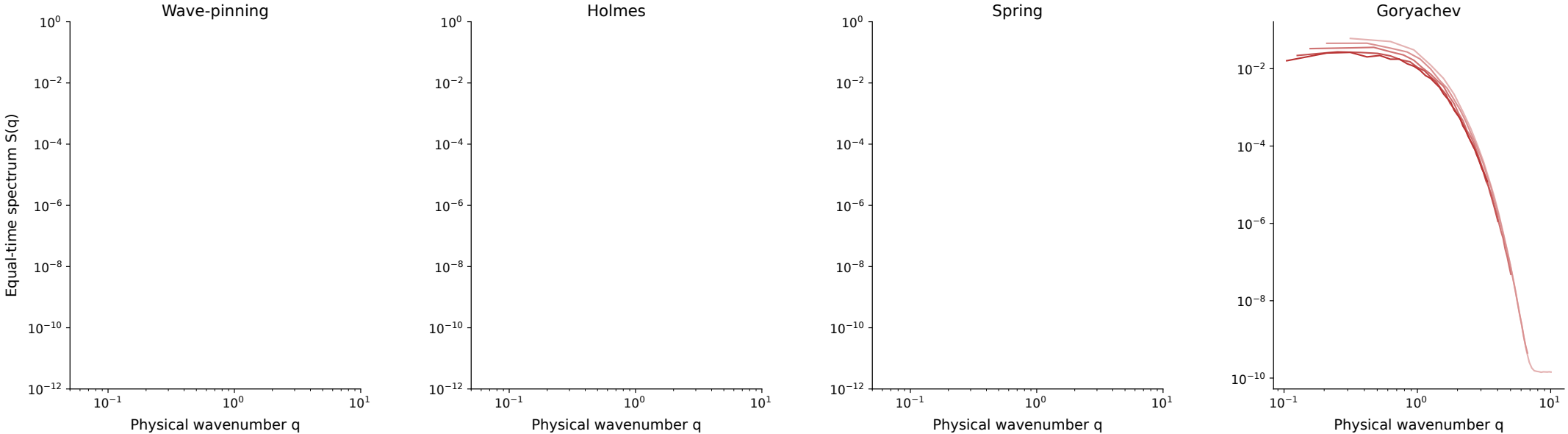
$S(q) = \langle |a_n|^2 \rangle / \langle a_0^2 \rangle$; n=1–32, two existing exemplars/condition. Spring q uses mean C. n* and C/n* are in exemplar_rates.csv.

S4.6 Equal-time spectrum, f = 0.02



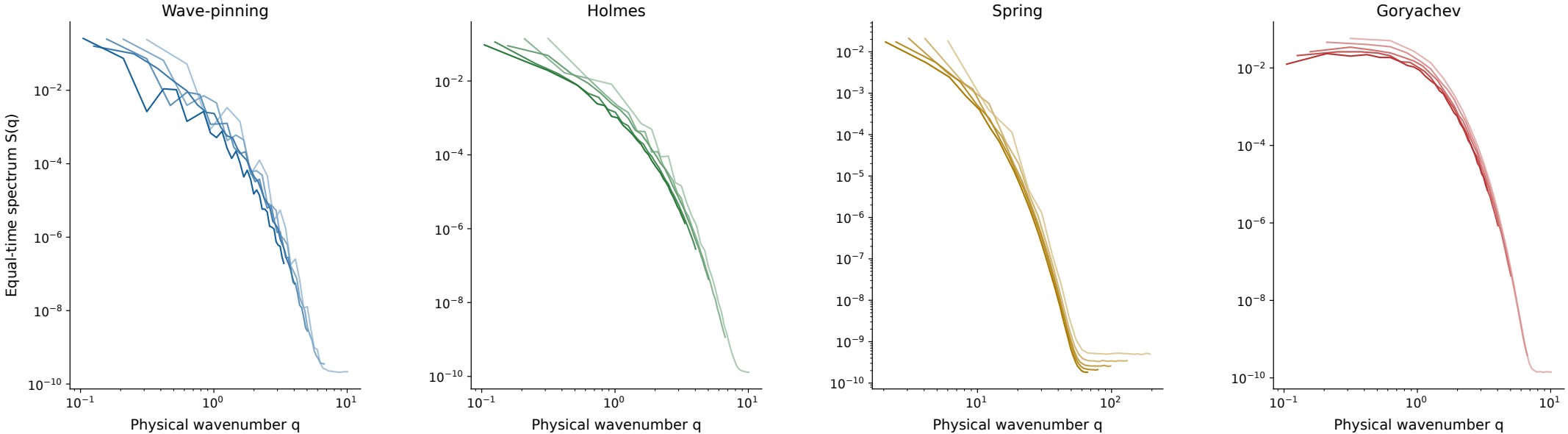
$S(q) = \langle |a_n|^2 \rangle / \langle a_0^2 \rangle$; n=1–32, two existing exemplars/condition. Spring q uses mean C. n* and C/n* are in exemplar_rates.csv.

S4.6 Equal-time spectrum, f = 0.025



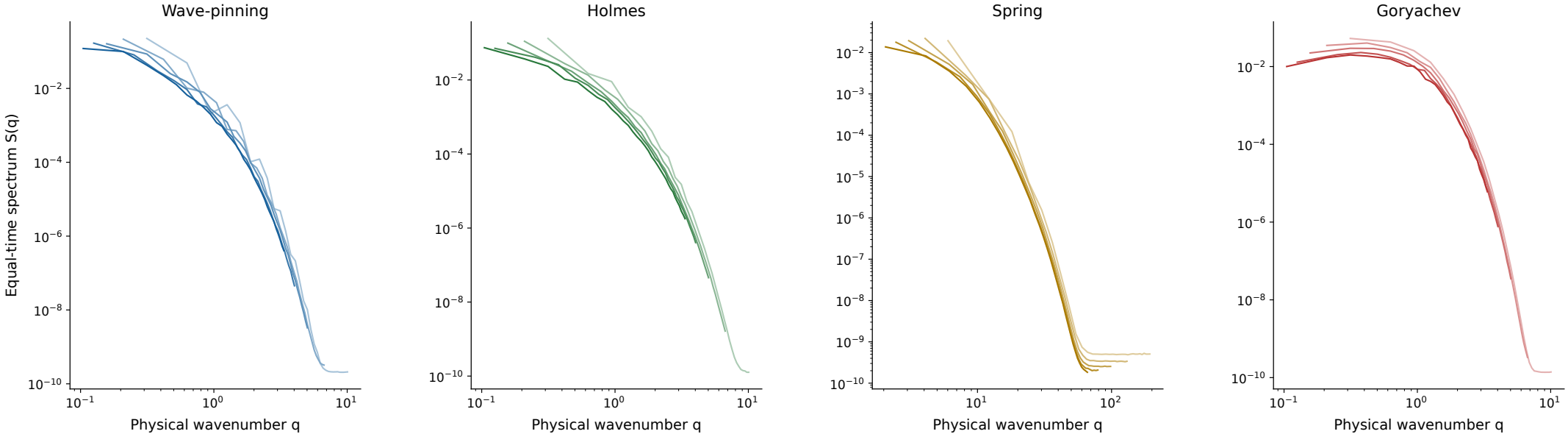
$S(q) = \langle |a_n|^2 \rangle / \langle a_0^2 \rangle$; n=1–32, two existing exemplars/condition. Spring q uses mean C. n* and C/n* are in exemplar_rates.csv.

S4.6 Equal-time spectrum, f = 0.03



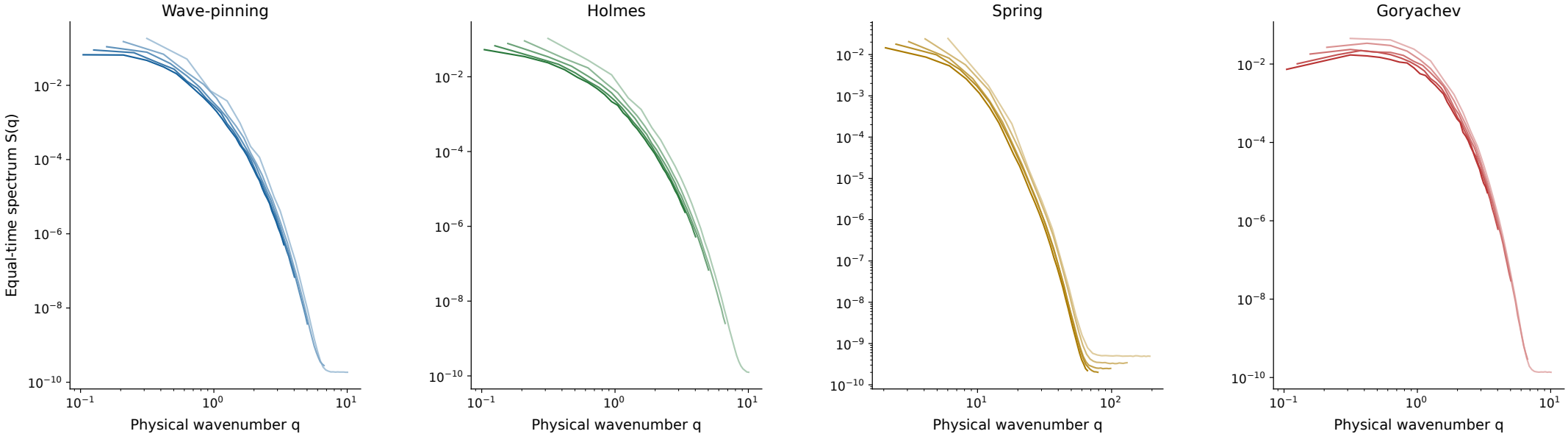
$S(q) = \langle |a_n|^2 \rangle / \langle a_0^2 \rangle$; n=1–32, two existing exemplars/condition. Spring q uses mean C. n* and C/n* are in exemplar_rates.csv.

S4.6 Equal-time spectrum, f = 0.05



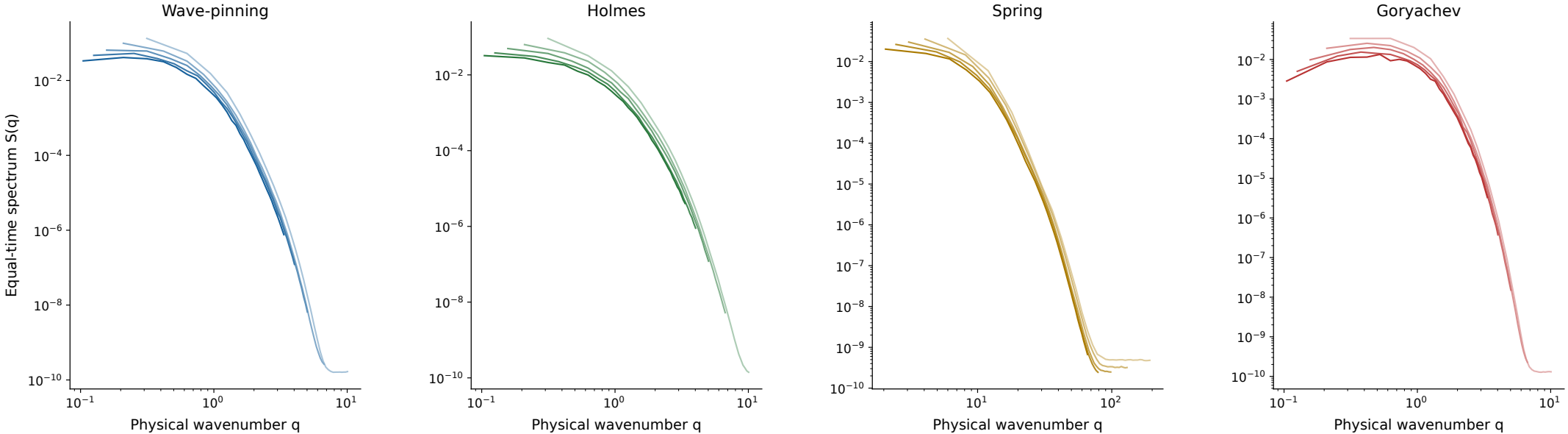
$S(q) = \langle |a_n|^2 \rangle / \langle a_0^2 \rangle$; n=1–32, two existing exemplars/condition. Spring q uses mean C. n* and C/n* are in exemplar_rates.csv.

S4.6 Equal-time spectrum, f = 0.1



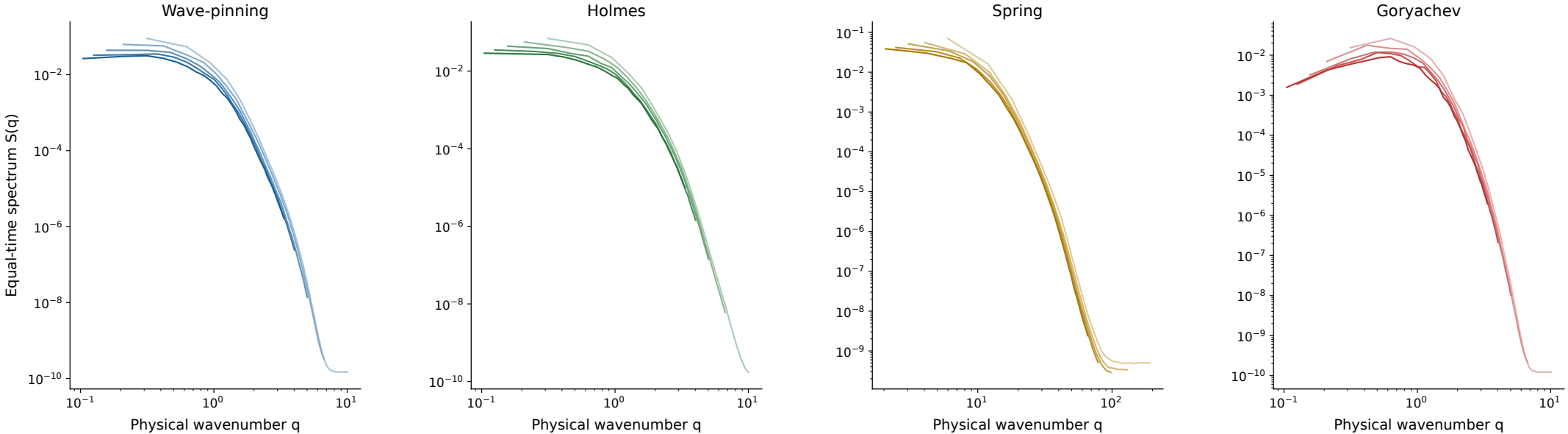
$S(q) = \langle |a_n|^2 \rangle / \langle a_0^2 \rangle$; n=1–32, two existing exemplars/condition. Spring q uses mean C. n* and C/n* are in exemplar_rates.csv.

S4.6 Equal-time spectrum, f = 0.3



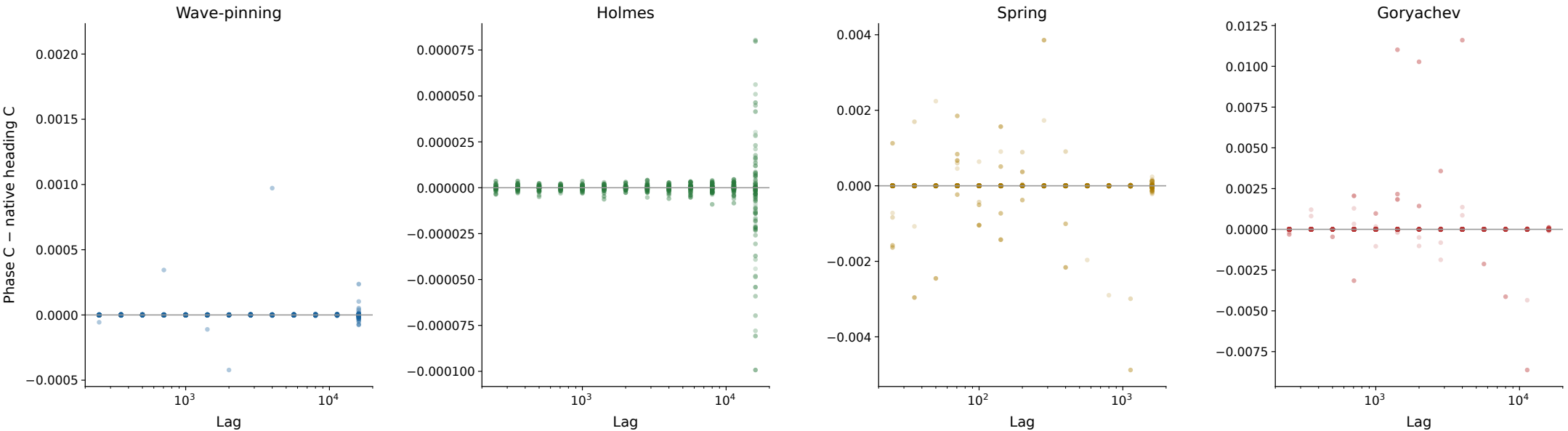
$S(q) = \langle |a_n|^2 \rangle / \langle a_0^2 \rangle$; n=1–32, two existing exemplars/condition. Spring q uses mean C. n* and C/n* are in exemplar_rates.csv.

S4.6 Equal-time spectrum, f = 1



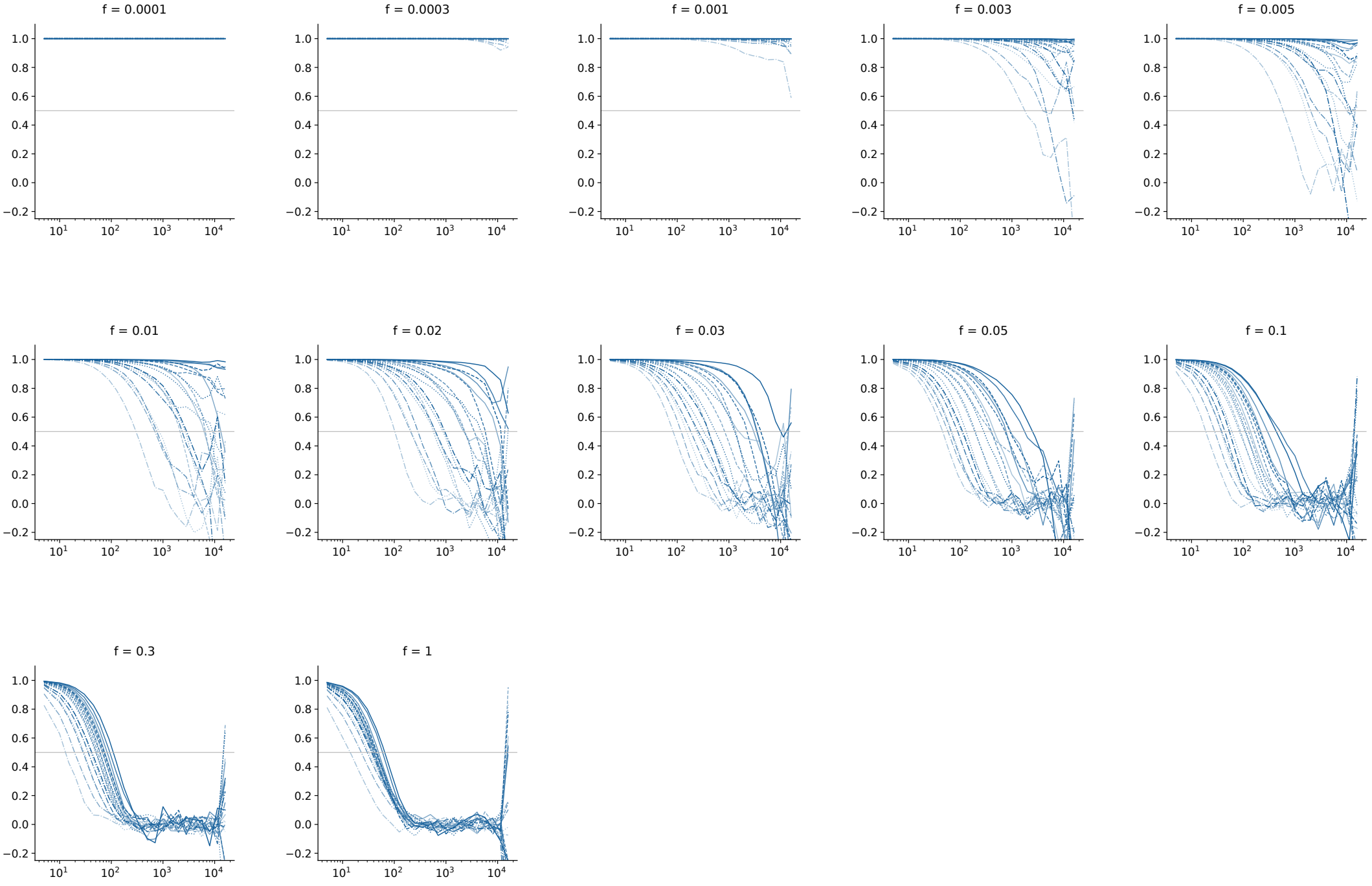
$S(q) = \langle |a_n|^2 \rangle / \langle a_0^2 \rangle$; n=1–32, two existing exemplars/condition. Spring q uses mean C. n* and C/n* are in exemplar_rates.csv.

S4.7 First-mode phase reproduces native heading correlation



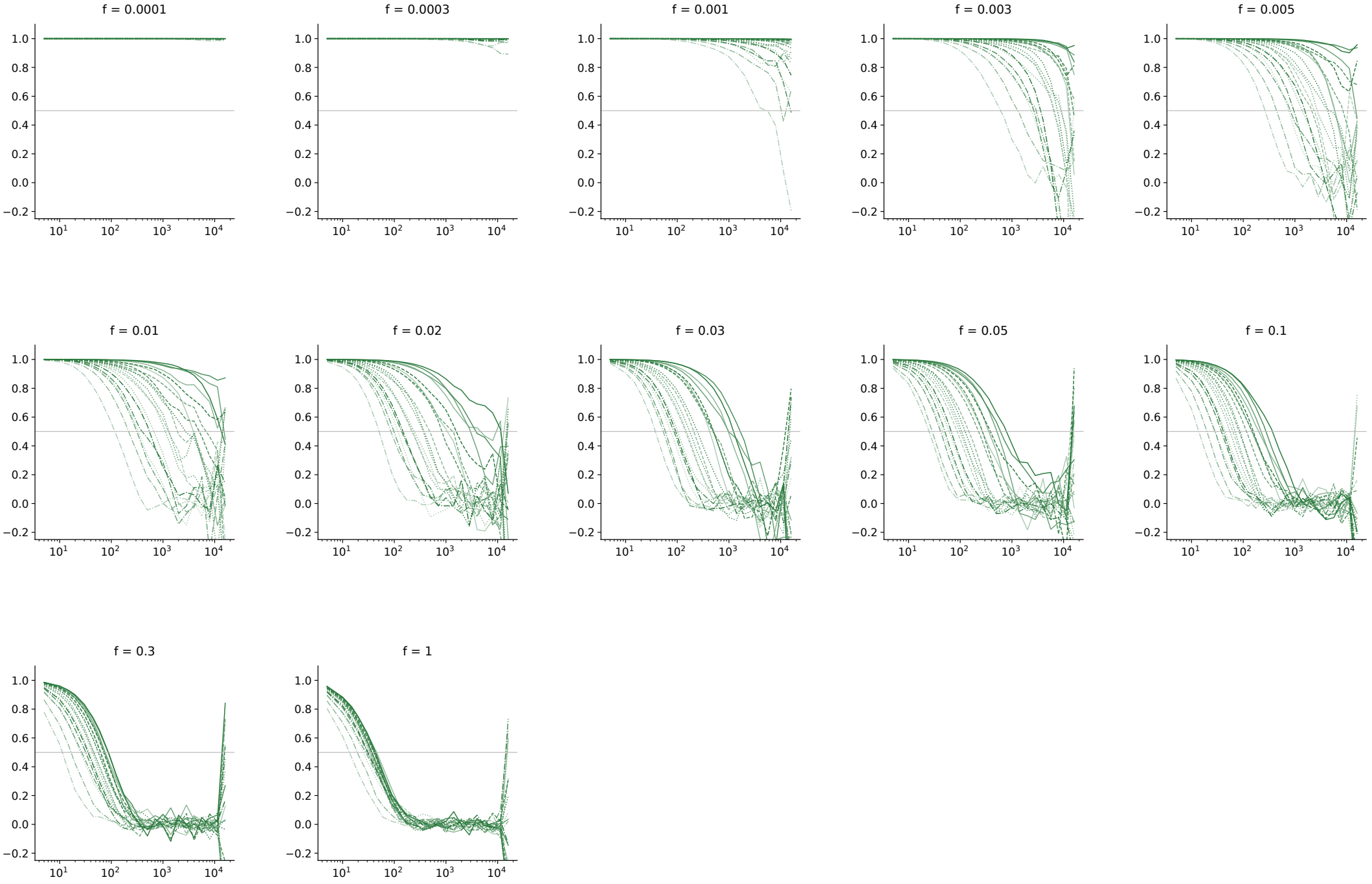
Difference between Fourier a1 phase correlation and native compact P heading, at identical exact saved endpoints. Eligibility uses each stored |P|≥0.1; nonzero differences include threshold crossing and bin/float16 precision.

S4.8 Normalized complex autocorrelations: Wave-pinning



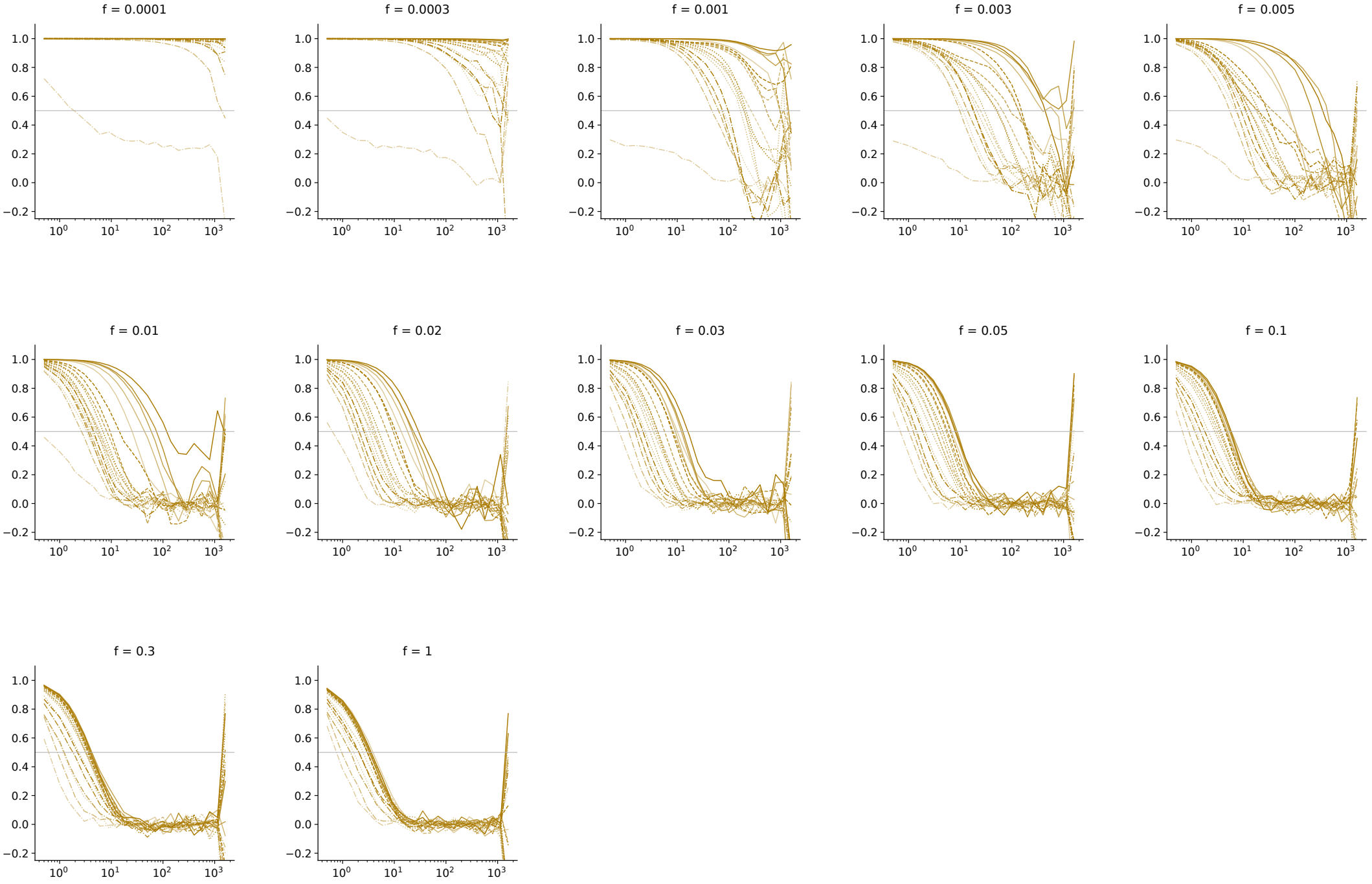
n=1 solid,2 dashed,4 dotted,8 dash-dot; size light→dark. All33 modes including n=0 are retained in compact CSV.

S4.8 Normalized complex autocorrelations: Holmes



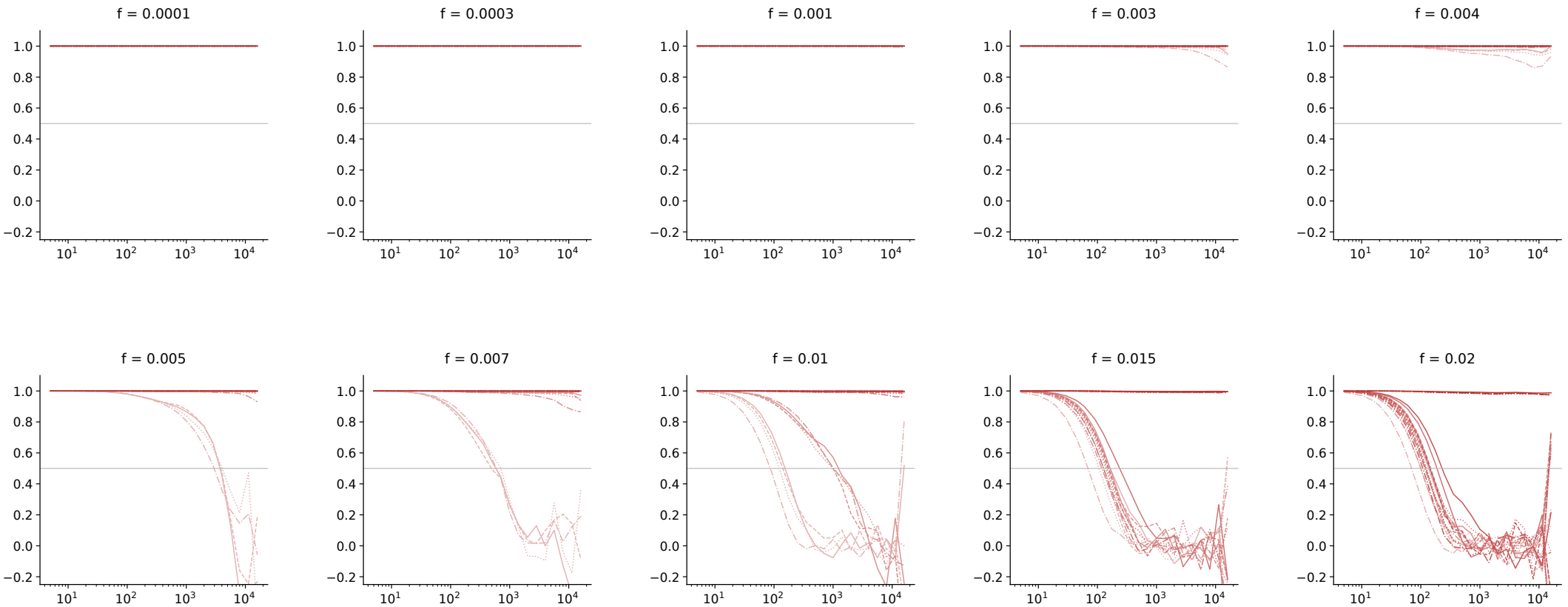
n=1 solid,2 dashed,4 dotted,8 dash-dot; size light→dark. All33 modes including n=0 are retained in compact CSV.

S4.8 Normalized complex autocorrelations: Spring



n=1 solid,2 dashed,4 dotted,8 dash-dot; size light→dark. All33 modes including n=0 are retained in compact CSV.

S4.8 Normalized complex autocorrelations: Goryachev



n=1 solid,2 dashed,4 dotted,8 dash-dot; size light→dark. All33 modes including n=0 are retained in compact CSV.